

An Open-Source Speech-to-Speech Pipeline with Lightweight Turn-Taking Classification for Code-Mixed English-French Speech

Rafiur Rahman Mashrafi
ID: 2221971042

Md. Shadman Shihab
ID: 2231639642

Md. Nafiul Alam Chowdhury
ID: 2231774642

CSE465 – Pattern Recognition and Neural Network

Abstract—This paper presents a fully open-source speech-to-speech dialogue pipeline that addresses three gaps in the recent DuplexCascade system [1]: reliance on proprietary components, English-only evaluation, and the need for large GPU clusters. Our pipeline chains Whisper-small for automatic speech recognition (ASR), a fine-tuned DistilBERT classifier for turn-taking decisions, TinyLlama-1.1B for response generation, and edge-tts for speech synthesis. The turn-taking classifier is trained on DailyDialog and predicts three actions: *wait*, *respond*, or *backchannel*. Training on a single free T4 GPU in approximately 45 minutes raises classifier accuracy from 88.48% (rule-based baseline) to 96.16%. We further compare classical pattern recognition methods – Gaussian Naive Bayes and k -Nearest Neighbors – against the neural classifier on the same task, connecting the work to core syllabus topics. The system runs entirely on free hardware and handles both English and code-mixed English-French audio.

Index Terms—speech-to-speech, turn-taking, DistilBERT, code-mixed, pattern recognition, Naive Bayes, k -NN

I. INTRODUCTION

Spoken dialogue systems need to decide, in real time, whether to keep listening, reply, or give a short acknowledgement. This decision is called *turn-taking*. Recent work like DuplexCascade [1] handles turn-taking by fine-tuning a 7-billion-parameter language model on 8 NVIDIA H100 GPUs over five hours. This makes the approach out of reach for most students and small research groups. The system also uses proprietary ASR and TTS components (DSM-ASR, DSM-TTS) and was never tested on code-mixed speech.

Our work asks a simpler question: can we build a speech-to-speech system that does turn-taking well, runs on a free GPU, uses only open-source tools, and works on code-mixed English-French speech?

We answer yes, with three contributions:

- 1) A complete open-source pipeline replacing all proprietary components.
- 2) A lightweight DistilBERT classifier (66M parameters) that learns turn-taking from text data and trains in 45 minutes on a free T4.
- 3) A comparison of classical pattern recognition methods (Gaussian Naive Bayes, k -NN, k -Means) against the neural approach on the same task.

II. RELATED WORK

DuplexCascade [1] is the paper we extend. It converts long utterances into short “micro-turns” and uses special tokens to control when the system speaks. Training requires 8 H100 GPUs and five hours. Moshi [2] is an end-to-end speech model that supports full-duplex interaction but loses the conversational intelligence of text-only LLMs. Our work stays in the cascaded design (ASR → classifier → LLM → TTS) but makes it lightweight and reproducible.

III. SYSTEM ARCHITECTURE

Our pipeline has four stages, shown in Fig. 1.

Speech → Whisper → DistilBERT → TinyLlama →
edge-tts → Speech

Fig. 1. Full speech-to-speech pipeline. DistilBERT is the only trained component. All others are pretrained and open-source.

Stage 1 – ASR. Whisper-small [3] transcribes the input audio and detects the language (English or French). It requires no fine-tuning and handles code-mixed input naturally.

Stage 2 – Turn-Taking. A fine-tuned DistilBERT classifier [4] reads the transcription and outputs one of three labels: *wait* (user is still speaking), *respond* (user finished, generate a reply), or *backchannel* (short acknowledgement like “uh-huh”). This replaces the 7B LLM used in DuplexCascade.

Stage 3 – Response Generation. TinyLlama-1.1B [5] generates a short reply only when the classifier outputs *respond*. We load it in 4-bit quantization using bitsandbytes to fit inside the T4’s 15 GB of VRAM.

Stage 4 – TTS. edge-tts converts the response text to speech. It selects an English or French voice based on the language detected in Stage 1.

IV. TURN-TAKING CLASSIFIER

A. Dataset and Labels

We use DailyDialog [6], a public English conversation dataset with 13,118 dialogues. We create labels automatically: utterances with no ending punctuation are labelled *wait*; complete utterances are *respond*; very short utterances matching a backchannel vocabulary (“yes”, “okay”, “I see”, etc.) are *backchannel*. This gives 51,184 training examples and 8,645 test examples.

B. Training

We fine-tune DistilBERT-base-uncased [4] with a three-way classification head. Because the *backchannel* class is rare, we apply class-weighted cross-entropy loss. Training runs for five epochs with learning rate 2×10^{-5} , batch size 32, and a linear warmup schedule. The whole training takes about 45 minutes on one free Colab T4 GPU.

C. Baseline

Before training, we build a rule-based classifier that uses the same labelling rules but no model. This gives us a before/after comparison that is fair and transparent.

V. EXPERIMENTS AND RESULTS

A. Classifier Accuracy

Table I shows the main result. The trained DistilBERT model reaches 96.16% accuracy on the test set, which is 7.68 percentage points above the rule-based baseline.

TABLE I
TURN-TAKING CLASSIFIER ACCURACY

Method	Accuracy	Train Time
Rule-based (baseline)	88.48%	None
DistilBERT (ours)	96.16%	~45 min

Table II shows training converged quickly. Best validation accuracy (96.16%) was reached at the first epoch; later epochs show mild overfitting as train accuracy rises but validation stays flat.

TABLE II
TRAINING HISTORY (5 EPOCHS)

Epoch	Tr. Loss	Tr. Acc	Val. Loss	Val. Acc
1	0.2942	94.23%	0.1526	96.16%
2	0.1509	96.22%	0.1201	96.12%
3	0.1141	96.73%	0.1309	95.47%
4	0.0834	97.44%	0.1528	95.08%
5	0.0608	98.28%	0.1746	95.26%

B. End-to-End Pipeline Evaluation

We test the full pipeline on six audio clips generated with edge-tts: three English and three code-mixed English-French. Each clip targets one label (*wait*, *respond*, *backchannel*). Table III shows the per-sample results.

Four out of six clips are correct (66.7%). Both failures are the *wait* class. The issue is that the TTS-generated audio clips end with natural sentence prosody, so Whisper adds a period to the transcript. The classifier then sees a complete sentence and correctly predicts *respond*. This is a known limitation of using synthetic audio for evaluation. With real human speech – where sentences genuinely trail off – we would expect better *wait* detection.

Table IV shows where the time goes. The biggest cost is TinyLlama inference (3.30s). Our total latency (6.22s) is higher

TABLE III
PER-SAMPLE PIPELINE RESULTS (AFTER TRAINING)

File	Exp.	Pred.	OK?	Latency
en_wait.wav	wait	respond	No	13.62s
en_respond.wav	respond	respond	Yes	8.85s
en_backchannel.wav	backchannel	backchannel	Yes	2.67s
fr_wait.wav	wait	respond	No	5.35s
fr_respond.wav	respond	respond	Yes	5.36s
fr_backchannel.wav	backchannel	backchannel	Yes	1.44s

than DuplexCascade (1.70s), but their system uses 8 H100 GPUs. The difference is hardware, not design.

TABLE IV
AVERAGE LATENCY BREAKDOWN

Component	Avg. Latency
ASR (Whisper-small)	1.604s
Classifier (DistilBERT)	0.068s
LLM (TinyLlama-1.1B)	3.297s
TTS (edge-tts)	1.248s
Total	6.218s
DuplexCascade (8×H100)	~1.70s

C. System Comparison

Table V compares our system against DuplexCascade on the dimensions that matter for reproducibility.

TABLE V
SYSTEM COMPARISON VS. DUPLEXCASCADE

Metric	DuplexCascade	Ours
Hardware	8× H100	1× T4 (free)
ASR	DSM-ASR*	Whisper-small
Turn-taking	Qwen2-7B*	DistilBERT
LLM	Qwen2-7B*	TinyLlama-1.1B
TTS	DSM-TTS*	edge-tts
Open-source	Partial	Full
Code-mixed	No	Yes (EN+FR)
Train time	5h / 8×H100	45min / 1×T4
Avg. latency	~1.70s	6.22s

* Proprietary, not publicly available

VI. CLASSICAL PATTERN RECOGNITION ANALYSIS

To connect this project to classical methods covered in the course, we compare two additional classifiers on the same turn-taking task.

A. Gaussian Naive Bayes

We represent each utterance as a TF-IDF vector (3,000 dimensions, unigrams and bigrams) and train a Gaussian Naive Bayes (GNB) classifier. GNB models the likelihood of each feature given a class as a Gaussian distribution and uses Bayes' theorem to compute the posterior:

$$P(\text{class} \mid \mathbf{x}) \propto P(\mathbf{x} \mid \text{class}) \cdot P(\text{class}). \quad (1)$$

Training takes under one second. GNB reaches 58.48% accuracy. The gap versus DistilBERT (96.16%) shows that the independence assumption between words does not hold for turn-taking – context across multiple words matters.

B. k -Nearest Neighbors

We extract the [CLS] token embedding (768 dimensions) from the pretrained DistilBERT encoder for every training and test utterance. We then apply k -Nearest Neighbors classification using Euclidean distance, which at $k=1$ is exactly the minimum distance classifier from the syllabus. Table VI shows the accuracy sweep.

TABLE VI
 k -NN ACCURACY ON DISTILBERT EMBEDDINGS

Method	Accuracy	Syllabus Topic
GNB (TF-IDF)	58.48%	Bayes classifier
k -NN, $k=1$	80.86%	Min. distance classif.
k -NN, $k=3$	82.70%	Min. distance classif.
k -NN, $k=5$	83.52%	Min. distance classif.
k -NN, $k=7$ (best)	84.13%	Min. distance classif.
DistilBERT (fine-tuned)	96.16%	Neural network

We also ran k -Means ($k=3$) on the embeddings and projected to 2D with PCA (25.7% variance explained). The Adjusted Rand Index (ARI) of 0.059 shows that the three classes do not form well-separated clusters in the raw embedding space – fine-tuning is what creates the separation.

The key takeaway is that representation quality matters more than classifier choice: k -NN on DistilBERT embeddings (84.13%) far outperforms GNB on TF-IDF (58.48%), even though k -NN is the simpler algorithm.

VII. DISCUSSION

What worked. The DistilBERT classifier trains quickly and achieves high accuracy on text data. Whisper correctly detects both English and French in code-mixed clips and picks the right TTS voice automatically. The whole system runs on hardware that is free for anyone.

What did not work as expected. The two *wait* clips were misclassified as *respond*. TTS-generated audio always sounds like complete sentences, so Whisper produces transcripts with periods, which the classifier reads as finished utterances. Real recordings of incomplete speech would likely fix this.

Latency gap. Our 6.22s average latency is higher than DuplexCascade’s 1.70s. TinyLlama inference alone costs 3.30s. The paper uses a 7B model on 8 H100s; we use a 1.1B model on one free T4. The gap is entirely explained by hardware.

VIII. CONCLUSION

We built a complete speech-to-speech turn-taking system using only free and open-source tools, trained on a single free T4 GPU in under an hour. The DistilBERT classifier improved turn-taking accuracy from 88.48% to 96.16% over a rule-based baseline. Comparing against Gaussian Naive Bayes (58.48%) and k -NN on embeddings (84.13%) shows clearly why fine-tuned neural representations outperform classical methods

for this task – but also where classical methods are useful (interpretability and speed). The system handles English and code-mixed English-French audio, extending DuplexCascade’s English-only scope without any proprietary components or large GPU clusters.

REFERENCES

- [1] J. Yang, Y. Fujita, and Y. Sudo, “DuplexCascade: Full-Duplex Speech-to-Speech Dialogue with VAD-Free Cascaded ASR–LLM–TTS Pipeline and Micro-Turn Optimization,” arXiv:2603.09180v1, 2026.
- [2] A. Défossez et al., “Moshi: A Speech-Text Foundation Model for Real-Time Dialogue,” Technical Report, 2024.
- [3] A. Radford et al., “Robust Speech Recognition via Large-Scale Weak Supervision,” in *Proc. ICML*, 2023.
- [4] V. Sanh, L. Debut, J. Chaumond, and T. Wolf, “DistilBERT, a Distilled Version of BERT: Smaller, Faster, Cheaper and Lighter,” arXiv:1910.01108, 2019.
- [5] P. Zhang et al., “TinyLlama: An Open-Source Small Language Model,” arXiv:2401.02385, 2024.
- [6] Y. Li et al., “DailyDialog: A Manually Labelled Multi-turn Dialogue Dataset,” in *Proc. IJCNLP*, 2017.